

DATA VORTEX — Round 1, Phase 1

Exploratory Data Analysis and Data Recovery Report

Theme: Rebuilding the Social Engine

1. Objective

Round 1 Phase 1 requires restoring the Social Engine intake pipeline from a deliberately corrupted social-media dataset. This report covers data cleaning, missing-value handling, anomaly identification, and exploratory analysis. Transformations are justified below. Missing fields are not invented. Values are computed from db1.csv (user profiles) and db2.csv (posts), matching the workflow in DataVortexEDAR1.ipynb.

2. Datasets

The recovery portal stored the intake data as CSV strings. After extraction, two tables were used.

Dataset	File	Role	Rows	Columns	Grain
Dataset 1 — posts	db2.csv	Post / content events	12,360 (raw)	8	One row per post record
Dataset 2 — users	db1.csv	Account profiles	1,500 (raw)	5	One row per user

Post fields: post_id, user_id, platform, text_content, timestamp, likes, shares, comments.

User fields: user_id, location, language, account_created, follower_count.

After duplicate removal, 12,000 unique posts remain. All 12,000 join to a user in Dataset 2 (join coverage 100.0%). Every one of the 1,500 users has at least one post (minimum 1, maximum 22, mean 8.00).

There are no orphan posts and no unused user profiles.

3. Data quality

3.1 Missing values — posts (raw, n = 12,360)

Column	Missing count	Missing %	Notes
post_id	0	0.00%	Complete identifier
user_id	0	0.00%	Complete foreign key
platform	1,846	14.94%	No secondary source to infer platform
text_content	1,746	14.13%	Text cannot be reconstructed
timestamp	0	0.00%	Present but mixed encodings
likes	1,858	15.03%	Separate from 525

			negative likes
shares	0	0.00%	Complete
comments	0	0.00%	Complete

3.2 Missing values — users (raw, n = 1,500)

Column	Missing count	Missing %
user_id	0	0.00%
location	0	0.00%
language	0	0.00%
account_created	0	0.00%
follower_count	0	0.00%

Inference: corruption is concentrated in the post table (platform, text, likes). The user table is structurally complete and can be used as a lookup without row drops.

4. Anomalies and recovery actions

Issue	Finding	Action	Justification
Duplicate post IDs	360 extra rows; 12,000 unique IDs of 12,360	Drop duplicate post_id, keep first	post_id is the record key; duplicates inflate volume and engagement
Conflicting duplicates	0 groups with differing content	None beyond de-duplication	The 360 extras are exact copies, not competing versions
Negative likes	525 rows (4.25% of raw posts)	Set likes to missing	Counts cannot be negative; a guessed replacement would fabricate data
Negative shares / comments	0 / 0	None	Those fields are valid as received
Missing platform	1,846 raw (14.94%); 1,784 after de-duplication	Keep as missing	Platform cannot be inferred from other fields without guessing
Missing text	1,746 raw (14.13%); 1,688 after de-duplication	Keep as missing	Post body cannot be safely reconstructed
Missing likes	1,858 raw (15.03%)	Keep as missing	Mean/median fill would create artificial engagement
Mixed timestamps	4,950 ISO; 3,788 Unix seconds; 3,622 DD-MM-YYYY; 0 other	Parse all three into one datetime	All three are valid encodings of the same field; conversion is reversible
HTML entities	341 posts	html.unescape	Formatting recovery, not new content
HTML tags	663 posts	Strip tags; convert to space	Removes markup without inventing words
UTF-8 / Latin-1 mojibake	316 posts	Repair where latin1→utf-8 round-trip is valid	Corrects encoding corruption already present in the string

4.1 Assumptions

1. post_id uniquely identifies a post. Duplicate IDs are repeated records, not distinct events.
2. Negative engagement is invalid, not a coded special value.
3. Missing platform, text, and likes have no defensible fill, so they stay missing.
4. Timestamp standardisation does not change the event; it only unifies representation.
5. HTML unescape, tag stripping, and reversible mojibake repair restore formatting rather than generate new data.
6. user_id is a stable join key between posts and profiles.

5. Cleaning outcome

Step	Before	After
Posts	12,360 rows	12,000 unique posts
Users	1,500 rows	1,500 rows (none removed)
Likes usable	10,502 non-missing, including negatives	9,677 non-negative likes
Likes missing (cleaned)	1,858	2,323 (original missing + 509 unique-post negatives)
Timestamp	Three mixed formats	YYYY-MM-DD HH:MM:SS; 0 unparsed
Join	—	12,000 / 12,000 matched (100%)

Cleaned exports from this pipeline: 12,000 posts, 1,500 users, and a 12,000-row merged table.

6. Statistical summary — posts

6.1 Engagement (cleaned unique posts unless noted)

Raw likes are distorted by negatives (minimum −4,987). The raw likes mean of 2,247.08 is not a usable location measure. After treating negatives as missing, likes are nearly symmetric (skew $\approx$ 0.005).

Metric	Likes (cleaned)	Shares (cleaned)	Comments (cleaned)
Count used	9,677	12,000	12,000
Missing	2,323	0	0
Mean	2,493.51	1,007.17	504.35
Std. deviation	1,438.29	575.07	288.68
Minimum	0	0	0
25th percentile	1,240	510	253
Median	2,502	1,018	503
75th percentile	3,724	1,501	755
Maximum	5,000	2,000	1,000
Skewness	0.005	−0.012 (raw)	−0.016 (raw)

Inference: likes, shares, and comments look like bounded uniform-style draws (caps at 5,000 / 2,000 / 1,000), not typical heavy-tailed social engagement. Means sit near the midpoints of those ranges. Zero engagement is rare (1 post with 0 likes among valid likes; 5 with 0 shares; 13 with 0 comments). Shares and comments can still be analysed on rows where likes are missing; dropping incomplete rows would discard 19.4% of unique posts unnecessarily.

6.2 Platform

Platform	Raw posts	Unique posts	Share of unique posts	Mean likes	Mean shares	Mean comments
Facebook	2,135	2,074	17.28%	2,526.55	984.17	506.94
YouTube	2,136	2,073	17.28%	2,528.33	1,011.83	504.38
Twitter	2,119	2,049	17.08%	2,439.87	1,005.39	506.13
Reddit	2,086	2,031	16.93%	2,487.80	1,002.23	511.18
Instagram	2,038	1,989	16.58%	2,494.51	1,040.84	499.80
Missing	1,846	1,784	14.87%	2,480.82	998.61	496.54

Inference: named platforms are nearly even (about 16.6–17.3% each). Missing platform is almost as large as any named platform (14.87%). Mean engagement does not differ in a practically large way across platforms (mean likes range 2,439.87–2,528.33). Platform is not a strong splitter of engagement in this intake.

6.3 Time

Parsed timestamps cover 1 May 2024 through 30 April 2025 inclusive (365 distinct dates; expected span 365 days; 0 gaps). Daily volume: mean 33.86 posts, median 34, minimum 18, maximum 59.

Month	Posts
2024-05	1,038
2024-06	979
2024-07	996
2024-08	1,015
2024-09	974
2024-10	1,029
2024-11	1,025
2024-12	1,035
2025-01	1,007
2025-02	914
2025-03	1,013
2025-04	975

Weekday	Posts	Mean likes
Monday	1,720	2,487.69
Tuesday	1,677	2,468.99
Wednesday	1,771	2,510.90
Thursday	1,718	2,479.43
Friday	1,723	2,473.39

Saturday	1,675	2,502.05
Sunday	1,716	2,531.63

Inference: posting is steady across the year. February 2025 is the lowest month (914), consistent with a shorter month rather than a collapse. Weekdays are close to even (1,675–1,771). There is no strong weekend or weekday engagement spike (mean likes 2,469–2,532). The intake looks like a continuous stream, not bursty campaigns.

6.4 Text

Among non-missing raw text: 10,520 posts contain hashtags and 1,415 contain @-mentions. After cleaning, the most frequent hashtags are evenly sized marketing tokens (Reviews 750, Fitness 749, BestValue 732, Eco 723, SpecialOffer 723, and similar counts down the list). That evenness, plus product/brand phrasing in sample posts, suggests generated review-style content rather than organic conversation.

7. Statistical summary — users

Metric	follower_count
Count	1,500
Missing	0
Mean	24,964.02
Std. deviation	14,185.48
Minimum	109
25th percentile	12,771.75
Median	24,741.50
75th percentile	37,097.25
Maximum	49,944
Skewness	0.016
Negative values	0

Account created dates run from 1 January 2023 to 31 December 2023. There are 10 language codes and 33 locations. Users by language: zh 168, ja 156, hi 156, en 153, fr 150, es 146, ru 144, ar 143, pt 143, de 141. The language mix is almost uniform (141–168). Locations are similarly spread; the largest cities are Shanghai and Barcelona (56 users each).

Country / territory (from location)	Users
USA	203
China	107
Germany	97
Italy	96
Japan	94
Brazil	93
Spain	92
India	92
UK	86
Canada	83

France	82
Australia	80
UAE	53
Singapore	49
South Africa	48
Mexico	40
Egypt	39
South Korea	36
Nigeria	30

Inference: follower_count is nearly symmetric around ~25,000 with a hard-looking cap near 50,000. That is unlike typical social graphs (usually right-skewed). Combined with even languages and even platforms, the user base looks synthetically balanced rather than organically concentrated.

8. User posting behaviour

Posts per user (cleaned)	Value
Users with ≥ 1 post	1,500 / 1,500
Mean	8.00
Std. deviation	2.85
Minimum	1
25th percentile	6
Median	8
75th percentile	10
Maximum	22
Share of posts from top 10 users	1.47%
Share from top 50 users	6.25%
Share from top 100 users	11.61%

Inference: activity is only mildly concentrated. The busiest user has 22 posts; the top 100 users write 11.61% of posts. This is not a power-user dominated corpus. Mean posts per user of exactly 8.00 on 12,000 posts and 1,500 users is consistent with a designed panel.

9. Cross-dataset analysis

9.1 Correlation (pairwise complete observations, cleaned merge)

	likes	shares	comments	follower_count
likes	1.000	-0.003	0.013	0.008
shares	-0.003	1.000	0.024	-0.014
comments	0.013	0.024	1.000	0.000
follower_count	0.008	-0.014	0.000	1.000

Inference: associations are essentially null. Followers do not track likes ($r = 0.008$). Likes, shares, and comments are also uncorrelated with each other ($|r| \leq 0.024$). In a real social engine, some positive

association is usual (reach vs engagement, or engagement metrics moving together). Here the metrics behave as independent columns. Correlation is descriptive only; it is not causation.

9.2 Mean engagement by follower quartile

Follower quartile	Mean likes	Mean shares	Mean comments
Q1 (lowest followers)	2,462.17	1,023.39	504.78
Q2	2,503.50	1,002.34	505.51
Q3	2,500.71	999.57	499.41
Q4 (highest followers)	2,508.16	1,003.37	507.68

Inference: moving from the lowest to highest follower quartile changes mean likes by only about 46 points on a ~2,500 scale. Audience size is not driving engagement in this dataset.

9.3 Mean likes by profile language

Language	Posts	Mean likes	Mean shares	Mean comments
en	1,220	2,571.37	997.11	504.18
ar	1,123	2,544.39	1,032.73	495.48
ja	1,298	2,509.30	995.97	510.10
zh	1,369	2,506.07	1,022.98	500.12
fr	1,156	2,498.82	980.25	519.76
pt	1,132	2,491.87	1,036.90	509.70
ru	1,119	2,484.44	991.12	512.72
de	1,149	2,472.33	1,003.82	498.80
es	1,143	2,459.46	1,032.32	488.74
hi	1,291	2,398.82	981.56	503.90

Highest location-level mean likes: Cairo 2,635.98; Barcelona 2,629.68; London 2,610.65. Lowest among locations checked: Lyon 2,325.58; Seoul 2,369.79; Johannesburg 2,389.94. The spread is modest relative to the 0–5,000 likes scale.

10. Findings for the restored intake

1. The post table was the corrupted intake: duplicates, mixed clocks, HTML/encoding residue, missing platform/text/likes, and invalid negative likes. The user table did not need row-level repair.
2. De-duplication (12,360 → 12,000) and converting 525 negative likes to missing restore logical consistency without fabricating replacements.
3. Time coverage is complete for one year (2024-05-01 to 2025-04-30) with stable daily volume. Mixed timestamp formats were a representation problem, not missing time.
4. Engagement and follower fields look independently generated and range-bounded. They should not be modelled as if follower count causes likes.
5. Platform, language, and location are balanced by design. Missing platform (~15%) is a real information gap and should stay coded as unknown in later SQL work, not redistributed.

6. Text is hashtag-heavy promotional/review copy with recoverable markup issues. Keep missing text as missing; do not impute captions.

11. Reproducibility

The notebook DataVortexEDAR1.ipynb loads both CSVs, profiles schema and missingness, counts duplicate IDs and negative likes, parses the three timestamp formats, cleans reversible text corruption, drops duplicate post_id rows, joins users on user_id, and writes cleaned CSVs. No analytical totals are hardcoded in that workflow. This report's numbers were recomputed from db1.csv and db2.csv with the same rules.

12. Rulebook compliance (Phase 1)

Requirement	How it is met
Cleaned dataset (CSV/JSON)	12,000 posts, 1,500 users, 12,000-row merge
EDA report	This document: quality, cleaning, statistics, inferences
Code notebook	DataVortexEDAR1.ipynb — cleaning, documentation, reproducible steps
Justify transformations	Section 4 actions and justifications
State assumptions	Section 4.1
No fabrication	No imputed platform, text, or likes; negatives set to missing, not replaced
Evaluation: cleaning, handling, EDA depth, standardisation, interpretation	Sections 3–10